

SOUMYA SINHA

+91 8282 855 660 | soumyasinha.ai@gmail.com | [Github](#) | [Tableau](#) | [Portfolio](#)

SUMMARY

Machine Learning Engineer with expertise in agentic AI, multimodal modeling, and explainable AI. With an interdisciplinary training I specialize in building complex AI systems bridging theoretical foundations to real-world deployment, while collaborating with diverse teams to advance high-impact scientific discovery.

EDUCATION

Master of Science in Information Management (Focus: Data Science)

Sept' 22 - Dec' 23

University of Washington | Seattle, WA

Courses Highlights : Data Science in R, Machine Learning in Python, Deep Learning LLM/LVLM, AI Ethics

Master's Thesis: Fake News Detection with Explainability (LIME & SHAP) |

- Designed and evaluated interpretable NLP pipelines comparing transformer-based, deep learning, and classical models for misinformation detection.
- Applied LIME and SHAP for feature attribution, enabling rigorous analysis of model reliability and decision transparency. Developed reproducible frameworks to assess robustness, generalization, and explainability across model architectures. Preprint: <https://doi.org/10.5281/zenodo.18925409>

Bachelor of Technology in Mathematics and Computing

Sept' 12 - Mar' 16

Delhi Technological University | New Delhi, India

Courses Highlights : Data Structures and Algorithms, OOP, Linear Algebra, Calculus, Network Analysis

SKILLS

Programming: Python, SQL, PyTorch, TensorFlow, PySpark

Development: AWS (S3, EC2, Lambda), Docker, CI/CD, Distributed Training (DDP/FSDP), Backend (FastAPI), Frontend (Streamlit), Vector DBs, LLM Frameworks (LangChain, LlamaIndex, CrewAI), SINDy, XAI (LIME, SHAP)

Soft Skills: Leadership, Cross-functional Collaboration, Research & Experimentation

EXPERIENCE

Data Scientist | SQL, PySpark, AWS (S3, EC2, Lambda), Machine Learning, NLP

New Jersey, U S

JerseySTEM

Feb' 24 -

present

- Built and deployed recommendation models to personalize learning content, increasing student engagement by ~22% and improving retention by ~15%.
- Developed scalable data pipelines and feature engineering workflows on 100K+ records, reducing processing latency by 40% and enabling real-time insights.
- Generated actionable insights from student behavior data, informing program strategy.

Development Scientist Intern | Python, SINDy, Physics-Informed Models

South San Francisco, US

Genentech

Jun' 23 - Sept'

23

- Created and implemented physics-informed machine learning procedures (SINDy) on pre-clinical PK/PD datasets to improve model interpretability and predict drug response on sparse experimental data.
- Presented a poster detailing the methodology and results at an internal Genentech conference, connecting cutting-edge computational techniques with insights from translational pharmacology.

Software Engineer | Python, Transformers, Distributed Training

Pune, India

Cerence

Aug' 20 - May'

22

- Led end-to-end development and deployment of an Indian-English conversational NLP/LLM system for in-vehicle voice commands, improving driving experience with accurate intent understanding and contextual responses (92% NER and intent accuracy).
- Architected scalable ML pipelines and model lifecycle workflows, ensuring reliability, latency optimization, and continuous model improvement in production environments.

- Optimized large-scale model training using DDP and FSDP, reducing training time by 45% and enabling efficient experimentation on production-scale datasets.

**Data Scientist | Python, Multi Omics Data, Machine Learning
CellWorks**

H.P., India
Sept' 16 – Nov'

19

- Developed machine learning models integrated with mechanistic biological frameworks to predict patient-specific drug responses, linking molecular alterations to phenotypic outcomes in cancer.
- Designed and analyzed multi-modal biomedical datasets, combining molecular, pharmacological, and pathway-level features to model complex disease mechanisms.
- Contributed to large-scale simulation of drug–target interactions and combination therapies, enabling exploration of therapeutic strategies across millions of potential treatment configurations.
- Applied statistical analysis and model evaluation techniques to refine predictions, ensuring consistency with underlying biological mechanisms and improving interpretability.

SELECTED PROJECTS

Agentic RAG Chatbot as Biology Tutor | CrewAI, LlamaIndex, ChromaDB, FastAPI, Streamlit, Docker, AWS EC2 | [github](#)

Built an agentic AI system for context-aware question answering on Class 12 Biology textbooks using a RAG pipeline with tool-driven retrieval and source-grounded responses, as a passion project to reinforce and revisit core science concepts. Designed a full-stack application with real-time chat, transparent reasoning, and scalable architecture, with plans to expand into Physics and Chemistry to create a comprehensive AI science tutor. Containerized and deployed on AWS EC2 using Docker for production-ready, end-to-end inference.

Proteomics-Based Cancer Signaling Network Analysis | Python, pandas, scikit-learn, NetworkX, g:Profiler

Analyzed publicly available cancer proteomics datasets to infer pathway activity and approximate signaling dynamics across tumor samples. Constructed protein interaction networks and performed pathway enrichment to identify key signaling modules linked to cancer subtypes. Explored immune-related patterns and potential therapeutic targets using statistical and graph-based methods, emphasizing biological interpretability.

Multi-Agent Stock Trading System | CrewAI, Groq, yfinance, Python | [github](#)

Built an agentic AI system that analyzes real-time stock data and generates Buy/Sell/Hold decisions using coordinated agents. Designed modular analyst and trader agents for structured reasoning and scalable decision-making.

Detection of Covid-19 from X-Ray scans | Pytorch, Computer Vision, Git | [github](#)

Implemented a multiclass image classification model to predict between Covid-19, pneumonia, and healthy lungs from chest X-Ray scans with an accuracy of 98% using ResNet-18 model.

Medical Image Analysis with Generative AI | Gemini-2.0, Vision-Language Models, Streamlit | [github](#)

Web Application to analyze user-submitted medical images, detect anomalies, and generate structured clinical reports with diagnostic insights and treatment recommendations; implemented safeguards to ensure reliable, unbiased, and ethically aligned outputs. [App link](#)

SELECTED ACHIEVEMENTS

- **UW Faculty Award for Student Excellence (2023):** Awarded to one outgoing student for academic & leadership excellence.
- **MSIM Dean's Scholarship Recipient (2022):** Awarded to top-ranking admits to the UW Information School.
- **First prize Meta's HerTech Hackathon (2018):** Developed solution to Make Smart Homes Safer for Women using voice recognition and geographical data. Recorded conversation is converted speech to text analyzed to generate actionable outputs based on the pattern of the users engagements, create customized intervention strategies and even alert the authorities, if need be.